

IMPLEMENTATION ROADMAP

CivicPulse

Version: 1.0

Methodology

Agile Scrum

Sprint Duration

1 Week

Total Sprint

9 Sprint

Sprint 0

Project Foundation

Goal

Membangun fondasi engineering.

Deliverables

- Repository Setup
- Monorepo Decision
- Next.js 16 Setup
- TypeScript Setup
- Tailwind Setup
- ESLint
- Prettier
- Husky
- Commitlint

Infrastructure

- Docker
- Docker Compose
- PostgreSQL
- Redis

Documentation

- PRD
- SDD
- DDD
- ADR

Success Criteria

Project dapat dijalankan lokal.

Sprint 1

Authentication & Authorization

Goal

Membangun sistem login.

Features

- Register
- Login
- Logout
- Session Management
- Role Management

Technical

- Better Auth
- Middleware
- Protected Routes

Success Criteria

RBAC berjalan.

Sprint 2

User & Agency Management

Goal

Membangun domain organisasi.

Features

- Create Agency
- Edit Agency
- Officer Management
- User Management

Success Criteria

Agency dan Officer dapat dikelola.

Sprint 3

Report Submission

Goal

Citizen dapat membuat laporan.

Features

- Create Report
- Upload Attachment
- Location Capture
- Validation

Technical

- Zod
- React Hook Form
- Upload Service

Success Criteria

Laporan berhasil dibuat.

Sprint 4

Report Workflow Engine

Goal

Workflow operasional berjalan.

Features

- Assign Report
- Change Status
- Comment System
- Report History

Workflow

SUBMITTED

↓

ASSIGNED

↓

IN_PROGRESS

↓

RESOLVED

↓

CLOSED

Success Criteria

Workflow berjalan penuh.

Sprint 5

Notification System

Goal

Komunikasi realtime.

Features

- In-App Notification
- Email Notification
- Notification Center

Technical

- Redis
- BullMQ

Success Criteria

Status update memicu notifikasi.

Sprint 6

Dashboard & Analytics

Goal

Visualisasi data.

Features

- Public Dashboard
- Officer Dashboard
- Admin Dashboard

Metrics

- Total Reports
- Resolution Rate
- Avg Resolution Time

Success Criteria

Analytics realtime tersedia.

Sprint 7

Audit & Security Hardening

Goal

Meningkatkan keamanan.

Features

- Audit Logs
- Rate Limiting
- Secure Headers

- Input Sanitization

Technical

- Arcjet
- Middleware Security

Success Criteria

Audit trail aktif.

Sprint 8

Production Readiness

Goal

Siap deploy.

Features

- Monitoring
- Logging
- Error Tracking
- Backup Strategy

Technical

- Sentry
- OpenTelemetry
- Pino

Success Criteria

Observability aktif.

Sprint 9

QA & Launch

Goal

Release Candidate.

Testing

- Unit Testing
- Integration Testing
- E2E Testing

Tools

- Vitest
- Testing Library
- Playwright

Deployment

- VPS
- Coolify
- Docker

Success Criteria

Production Release v1.0

Go Live.